

Math 340: Elementary Matrix and Linear Algebra

MIDTERM ONE

Thursday, October 10th, 2024, 7:30pm-9:00pm

Circle your Instructor and your TA:

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I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of eight problems for a total of 50 points; the exam is seven sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam packet will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.
- **Notation:** $\mathbf{0}$ denotes the zero vector, O denotes the zero matrix, and I_n denotes the $n \times n$ identity matrix.

DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.

1. (8 points, 1 point each) Clearly mark the correct answer(s) for each of the following by completely filling in the appropriate bubble. **No justification is needed.**

a.) (True/False) $\begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is a row echelon matrix.

- ☒ True
☐ False

- b.) (True/False) If A is an 3×3 matrix such that $A^2 = O$, then $A = O$.

- ☐ True
☒ False

ex: $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

- c.) (True/False) If two square matrices A and B are row equivalent, then $\det(A) = \det(B)$.

- ☐ True
☒ False

ex: $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{2r_1 \rightarrow r_1} B = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$
 $\det(A) = 1$ $\det(B) = 2$

- d.) (True/False) If A is a 3×3 matrix with $\det(A) = 6$, then $\det(3A) = 18$.

- ☐ True
☒ False

||
 $3^3 \cdot 6 = \text{big ol' number that isn't } 18.$

e.) (True/False) $\begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 4 \end{bmatrix}$ are row equivalent.

- ☐ True
☒ False

rank 2 rank 3.

f.) **(Multiple Choice-choose one)** Suppose A is a 5×5 matrix. If the rank of A is 3, then the homogeneous system $A\mathbf{x} = \mathbf{0}$ will have...

- ☐ No solutions.
- ☐ Exactly one solution.
- ☐ Exactly two solutions.
- ☐ Exactly three solutions.
- ☒ Infinitely many solutions.

↓
2 free variables.

g.) **(Multiple Choice-choose one)** If A is 2×3 , B is 4×4 , and C is 4×3 , then what is the size of the matrix AC^TB ?

- ☐ 4×4
- ☐ 2×3
- ☐ 3×4
- ☐ 4×3
- ☒ None of the above.

$2 \times 3 \quad 3 \times 4 \quad 4 \times 4 \rightarrow 2 \times 4$ "

h.) **(Multiple Choice-choose one)** Suppose A is a 3×3 skew-symmetric matrix. Consider the following statements:

- a. $A + A^T = O$
- b. $A^T - A = O$
- c. $A + I_3 = I_3 + A$
- d. $A\mathbf{x} = -A^T\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^3$.
- e. $A\mathbf{x} = \mathbf{0}$ for all $\mathbf{x}$ in $\mathbb{R}^3$.

$$A^T = -A.$$

Which of these statements is true about A ?

- ☐ (a), (b), and (d)
- ☒ (b), (c), and (d)
- ☐ (b) and (c)
- ☐ (a) and (e)
- ☐ None of the above.

2. (7 points total)

- a. (3 points) Suppose A, B are 4×4 matrices, and B is obtained from A by the following elementary row operations:

Multiply r_3 by $\frac{1}{6}$

Add $7r_1$ to r_2 (and replace r_2)

Swap r_2 and r_4

Add $-r_3$ to r_4 (and replace r_4)

Multiply r_3 by 10

$$A \rightarrow B$$

If $\det(B) = -4$, find $\det(A)$. Show all work.

$$A \xrightarrow{\frac{1}{6}r_3} A_1 \xrightarrow{7r_1+r_2 \rightarrow r_2} A_2 \xrightarrow{r_2 \leftrightarrow r_4} A_3 \xrightarrow{-r_3+r_4 \rightarrow r_4} A_4 \xrightarrow{10r_3 \rightarrow r_3} A_5 = B$$

$$\xrightarrow{\frac{1}{6}\det(A)} \xrightarrow{\frac{1}{6}\det(A)} \xrightarrow{-\frac{1}{6}\det(A)} \xrightarrow{-\frac{1}{6}\det(A)} \xrightarrow{-\frac{10}{6}\det(A)} \det(B) = -4$$

$$\det(A) = \boxed{\frac{+4 \cdot 6}{10} = \frac{24}{10}}$$

- b. (4 points) Suppose C, D, E are 3×3 matrices, where $\det(C) = 5$ and $\det(D) = -2$. If $C^{-1}D^T E = I_3$, find $\det(E)$. Show all work.

$$\downarrow$$

$$\det(C^{-1}D^T E) = \det(I_3) = 1$$

$$\det(C^{-1}) \det(D^T) \det(E) = 1$$

$$\frac{1}{5}(-2) \det(E) = 1 \Rightarrow \boxed{\det(E) = \frac{-5}{2}}$$

3. (6 points) Let $A = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$.

a. (3 points) Find all the eigenvalues of A . Show all work. (Hint: One of the eigenvalues is 5).

$$\begin{aligned} \det(\lambda I - A) &= \det \begin{bmatrix} \lambda - 3 & -1 \\ -2 & \lambda - 4 \end{bmatrix} = (\lambda - 3)(\lambda - 4) - 2 = 0 \\ &= \lambda^2 - 7\lambda + 12 - 2 = 0 \\ &= \lambda^2 - 7\lambda + 10 = 0 \\ &= (\lambda - 5)(\lambda - 2) = 0 \\ &\quad \boxed{\lambda = 5, 2} \end{aligned}$$

b. (3 points) Find an eigenvector of A corresponding to $\lambda = 5$. Show all work.

$$5I - A = \begin{bmatrix} 5-3 & -1 \\ -2 & 5-4 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -2 & 1 \end{bmatrix} \xrightarrow{r_1 + r_2 \rightarrow r_2} \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2v_1 - v_2 = 0$$

$$\text{Let } v_1 = 1$$

$$2 = v_2$$

$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigenvector
for A
corresponding to
 $\lambda = 5$.

4. (6 points total) Suppose A, B, C, D are 2×2 matrices, where $D^{-1} = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix}$, $A = \begin{bmatrix} 1 & -3 \\ 2 & 2 \end{bmatrix}$, and $D(B+C) = A$.

- a. (2 point) Which of the following is a correct expression for C ? (Choose one. No justification is required).

☐ $C = AD^{-1} - B$

☒ $C = D^{-1}A - B$

$$B+C = D^{-1}A \rightarrow C = D^{-1}A - B$$

- b. (4 points) Using your answer from part (a), calculate matrix C . Show all work.

correct $C = D^{-1}A - B$

$$C = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ 2 & 2 \end{bmatrix} - \begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & -8 \\ 5 & -7 \end{bmatrix} - \begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix}$$

$$C = \begin{bmatrix} 3 & -10 \\ 4 & -6 \end{bmatrix}$$

incorrect $C = AD^{-1} - B$

$$C = \begin{bmatrix} 1 & -3 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix}$$

$$C = \begin{bmatrix} -7 & -4 \\ 10 & 0 \end{bmatrix} - \begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix}$$

$$C = \begin{bmatrix} -4 & -6 \\ 9 & 1 \end{bmatrix}$$

5. (6 points total) Suppose a linear system of 3 equations and 3 unknowns (x, y, z) reduces to the following form via elementary row operations:

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & (k-3) & (k+1) \end{array} \right]$$

- (a) (2 points) For what value(s) of k (if any) will the system have **no solutions**? Briefly explain your answer.

If $k=3$, last row becomes:

$[0 \ 0 \ 0 : 4]$ $0 \neq 4$, which means the system is inconsistent.

- (b) (4 points) If $k = 5$, finish reducing the augmented matrix to **reduced row echelon form**, then solve for x, y, z . Show all work.

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 2 & 6 \end{array} \right] & \xrightarrow{\frac{1}{2}r_3 \rightarrow r_3} \left[\begin{array}{ccc|c} 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] & \xrightarrow{-2r_3 + r_2} \left[\begin{array}{ccc|c} 1 & 3 & 0 & 2 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 3 \end{array} \right] \\ & \xrightarrow{-3r_2 + r_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 14 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 3 \end{array} \right] \\ & \boxed{\begin{array}{l} x=14 \\ y=-4 \\ z=3 \end{array}} \end{aligned}$$

6. (8 points total) (this problem has overflow room on the next page)

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 \\ 2x_2 \\ 2x_1 - x_2 \end{bmatrix}$.

a. (2 points) Evaluate $T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right)$.

$$\begin{array}{c} 11 \\ \left[\begin{array}{c} 2+2\cdot 3 \\ 2\cdot 3 \\ 2\cdot 2-3 \end{array} \right] = \left[\begin{array}{c} 8 \\ 6 \\ 1 \end{array} \right] \end{array}$$

b. (2 points) Find the matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \\ 2 & -1 \end{bmatrix}$$

c. (4 points) Is there an $\mathbf{x}$ in $\mathbb{R}^2$ such that $T(\mathbf{x}) = \begin{bmatrix} -2 \\ -6 \\ 11 \end{bmatrix}$? Fully justify your answer.

solve:

$$\left[\begin{array}{cc|c} 1 & 2 & -2 \\ 0 & 2 & -6 \\ 2 & -1 & 11 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array} \right]$$

Yes, system is consistent:
let $x_1 = 4, x_2 = -3$

easy: middle says $2x_2 = -6$
 $x_2 = -3$

top says: $x_1 + 2(-3) = -2$

$$x_1 = -2 + 6 = 4$$

bottom says: $2(4) - (-3) = 8 + 3 = 11$

Yes! use $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ -3 \end{bmatrix}$

(Overflow room for Problem 6, if needed)

Don't need this.



7. (5 points) Find A , if $(I_2 + 4A)^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & -1 \end{bmatrix}$. Show all work.

↓ take inverse of both sides

$$I_2 + 4A = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \quad \text{check:} \quad \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$4A = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -3 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 0 \\ -3/4 & 0 \end{bmatrix}$$

8. (4 points) For which value(s) of c does $A = \begin{bmatrix} 3 & 0 & 1 \\ -c & 0 & 3 \\ -1 & 2 & 1 \end{bmatrix}$ have a **nontrivial** solution for $A\mathbf{x} = \mathbf{0}$?

Show all work, and justify your answer.

Use det:

$$\det A = -2(9-c) = 0$$

$$c = 9$$

If $c=9$, $\det(A)=0$,
so A is not invertible,
meaning it has a
nontrivial solution to

$$A\vec{x} = \vec{0} \quad (\text{Box 0' Facts})$$

Row reduce A :

$$\begin{bmatrix} 3 & 0 & 1 \\ c & 0 & 3 \\ -1 & 2 & 1 \end{bmatrix} \xrightarrow{-r_3, r_1 \leftrightarrow r_3} \begin{bmatrix} 1 & -2 & -1 \\ c & 0 & 3 \\ 3 & 0 & 1 \end{bmatrix}$$

$$\downarrow \begin{array}{l} -cr_1 + r_2 \rightarrow r_2 \\ -3r_1 + r_3 \rightarrow r_3 \end{array}$$

$$\begin{bmatrix} 1 & -2 & -1 \\ 0 & 2c & 3+c \\ 0 & 6 & 4 \end{bmatrix}$$

$$\downarrow \frac{1}{6}r_3, r_2 \leftrightarrow r_3$$

$$\begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & \frac{2}{3} \\ 0 & 2c & 3+c \end{bmatrix}$$

$$\downarrow -2cr_2 + r_3$$

$$\begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & \frac{2}{3} \\ 0 & 0 & -\frac{4c}{3} + 3+c \end{bmatrix}$$

$$-\frac{4c}{3} + 3 + c = 0$$

$$-\frac{1}{3}c + 3 = 0$$

$$\boxed{c = 3 \cdot 3 = 9}$$

If $c=9$, we get a row
of zeros $\Rightarrow$ free variable
 $\Rightarrow$ nontrivial solution to
 $A\vec{x} = \vec{0}$.